Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Formulates correctly one non-trivial inequality	AO3.1b	B1	$3x + 2y + z \le 360$
	Formulates correctly a second non-trivial inequality	AO3.1b	B1	$40x + 20y + 5z \le 2500$
	States correctly all three trivial inequalities	AO1.2	B1	$x \ge 0,\ y \ge 0,\ z \ge 0$

(b)(i)	Translates 'their' inequalities into a simplex tableau with two slack variables	AO3.1a	M1	See diagram on the next page
	Identifies a correct pivot from 'their' tableau	AO1.1b	A1	
	Uses the simplex algorithm correctly to modify the 'their' two non-pivot rows	AO1.1a	M1	
	Identifies the correct pivot in 'their' modified tableau	AO1.1b	A1F	
	Uses the simplex algorithm correctly to modify 'their' two non-pivot rows	AO1.1a	M1	
	Makes a correct interpretation of 'their' final tableau in the context of the problem, provided the objective row is non-negative	AO3.2a	E1	To maximise the profit, the company should repair and sell 28 monitors, 0 hard drives and 276 keyboards each month
(ii)	States no further use of the simplex algorithm is required and explains why	AO2.4	R1	The objective row of the final tableau being non-negative numbers.

P	x	y	z	s	t	value
1	-80	-35	-15	0	0	0
0	3	2	1	1	0	360
0	<u>40</u>	20	5	0	1	2500
1	0	5	-5	0	2	5000
0	0	0.5	<u>0.625</u>	1	-0.075	172.5
0	1	0.5	0.125	0	0.025	62.5
1	0	9	0	8	1.4	6380
0	0	0.8	1	1.6	-0.12	276
0	1	0.4	0	-0.2	0.04	28

(iii)	Introduces a new inequality for the hard drive hardware, ensuring that at least some hard drives are required to be repaired and sold	AO3.4	E1	As $y = 0$, enforce some hard drives to be repaired by requiring that, for instance, $y \geq 10$
				Total 11 marks

Q2.

(a)

P	x	y	z	r	s	t	value
1	–1	2	–3	0	0	0	0
0	1	1	1	1	0	0	16
0	1	–2	2	0	1	0	17
0	2	–1	2	0	0	1	19

All correct, 3 rows correct

B2,1,0 2

(b) (i) z-col: $\frac{16}{1}, \frac{17}{2}, \frac{19}{2}$

M1

Min, R_3 as pivot

A1 2

(ii)

1 $\frac{1}{2}$ –1 0 0 $1\frac{1}{2}$ 0 $\frac{51}{2}$

$$\begin{array}{cccccccc} 0 & \frac{1}{2} & 2 & 0 & 1 & -\frac{1}{2} & 0 & \frac{15}{2} \\ 0 & \frac{1}{2} & -1 & 1 & 0 & \frac{1}{2} & 0 & \frac{17}{2} \\ 0 & 1 & 1 & 0 & 0 & -1 & 1 & 2 \end{array}$$

Row operations

M1

One row (other than R_3) correct

A1

All correct

A1

3

Alternative

$$\left(\begin{array}{ccccccc|c} 2 & 1 & -2 & 0 & 0 & 3 & 0 & 51 \\ 0 & 1 & 4 & 0 & 2 & -1 & 0 & 15 \\ 0 & 1 & -2 & 2 & 0 & 1 & 0 & 17 \\ 0 & 1 & 1 & 0 & 0 & -1 & 1 & 2 \end{array}\right)$$

(M1)

(A1)

(A1)

(c) (i) y col $\frac{15}{4}$, $\left(-\frac{17}{2}\right)$, $\frac{2}{1}$ R_4 as pivot

Fully correct description

B1

$$\begin{array}{cccccccc} 1 & 1\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & \frac{55}{2} \\ 1 & 1\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & \frac{55}{2} \\ 0 & -1\frac{1}{2} & 0 & 0 & 1 & 1\frac{1}{2} & -2 & \frac{7}{2} \end{array}$$

0	$1\frac{1}{2}$	0	1	0	$-\frac{1}{2}$	1	$\frac{21}{2}$
0	1	1	0	0	–1	1	2

Row operations

M1

All correct

A1

Alternative

2	3	0	0	0	1	2	55
0	–3	0	0	2	3	–4	7
0	3	0	2	0	–1	2	21
0	1	1	0	0	–1	1	2

M1

A1

(ii) Optimal

$P = \frac{55}{2}$

Both statement and value needed. OE

B1

$x = 0,\ y = 2,\ z = \frac{21}{2}$

B1

$s = t = 0,\ r = \frac{7}{2}$

B1

3

[13]

Q3.

(a)

P	x	y	z	r	s	t	value

$$\begin{array}{ccccccccc} 1 & -4 & -3 & -1 & 0 & 0 & 0 & & 0 \\ 0 & 2 & 1 & 1 & 1 & 0 & 0 & & 25 \\ 0 & 1 & 2 & 1 & 0 & 1 & 0 & & 40 \\ 0 & 1 & 1 & 2 & 0 & 0 & 1 & & 30 \end{array}$$

All correct, 3 rows correct

B2,1,0

2

(b)

$$\begin{array}{cccccccc} 1 & 0 & -1 & 1 & 2 & 0 & 0 & 50 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{25}{2} \\ 0 & 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 & \frac{55}{2} \\ 0 & 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 & \frac{35}{2} \end{array}$$

Pivot, x-col: 12.5, 40, 30 seen and correct pivot chosen

B1

Row operations

M1

All correct

A1

3

(c) (i)

$$\begin{array}{cccccccc} 1 & 0 & 0 & \frac{4}{3} & \frac{5}{3} & \frac{2}{3} & 0 & \frac{205}{3} \\ 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & 0 & \frac{10}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 & \frac{55}{3} \\ 0 & 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 & \frac{25}{3} \end{array}$$

Pivot, y-col: their 25, 55 / 3, 35 seen and correct pivot chosen

B1

Row operations

M1

All correct

A1

3

(ii) Max $P = \frac{205}{3}$

Condone optimal, etc

B1

$x = \frac{10}{3}$, $y = \frac{55}{3}$, $z = 0$

Ft on x and y

B1

$r = 0$, $s = 0$, $t = \frac{25}{3}$

All 3 must be stated

B1ft

3

Alternative

Comments as above

(a)

P	x	y	z	r	s	t	value
1	–4	–3	–1	0	0	0	0
0	2	1	1	1	0	0	25
0	1	2	1	0	1	0	40
0	1	1	2	0	0	1	30

(2)

(b)

1	0	–1	1	2	0	0	50
0	2	1	1	1	0	0	25
0	0	3	1	–1	2	0	55

0	0	1	3	–1	0	2	35

(3)

(c) (i)

3	0	0	4	5	2	0	205
0	6	0	2	4	–2	0	20
0	0	3	1	–1	2	0	55
0	0	0	8	–2	–2	6	50

(3)

(ii) $P = \frac{205}{3}$

$x = \frac{10}{3}$, $y = \frac{55}{3}$, $z = 0$

$r = s = 0$, $t = \frac{25}{3}$

(3)

[11]

Q4.

(a) (i) x-column

B1

pivot = 6

B1

$\frac{2}{2} = 1$, $\frac{3}{6} = \frac{1}{2}$ $\left(\text{and } \frac{1}{2} < 1\right)$
smallest **positive** quotient

need to see correct quotients considered
*negative value **must** be mentioned*
as being considered but rejected

E1

3

(ii)

P	x	y	z	s	t	u	**value**
1	0	1	0	1	$\frac{1}{3}$	0	7

0	0	13	1	3	$-\frac{1}{3}$	0	1
0	1	–5	0	–1	$\frac{1}{6}$	0	$\frac{1}{2}$
0	0	–14	0	–4	$\frac{1}{6}$	1	$4\frac{1}{2}$

row operations M1

1st, 2nd or 4th row correct A1

another of these 3 correct A1

all correct (condone multiples of rows) A1 **4**

(b) (i) No negatives in **top row**

but *must have no negative values in "their" top row* E1 **1**

(ii) One (inequality still has slack) B1 **1**

(c) (i) $P = 7$

FT their tableau B1 √

$x = \frac{1}{2}$, $y = 0$, $z = 1$

condone one slip in final tableau B1 cao **2**

(ii) Substituting "their" values from (c)(i)

$\frac{1}{2}k + 0 + 3 = 7$

M1

$\Rightarrow k = 8$

A1

2

[13]

Q5.

(a)

P	x	y	z	s	t	value
1	$-k$	-6	-5	0	0	0
0	2	1	4	1	0	11
0	1	3	6	0	1	18

may have 1's in 's' and 't' columns interchanged
second row correct

B1

third row correct

B1

2

(b)

1	$2-k$	0	7	0	2	36
0	$\frac{5}{3}$	0	2	1	$-\frac{1}{3}$	5
0	$\frac{1}{3}$	1	2	0	$\frac{1}{3}$	6

may earn next B1 M1 if no slack variables
pivot is 3 (identified or used)

B1

row operations (even with wrong pivot)
(obtaining 0 in pivot column)

M1

first or second row correct

A1

all correct (condone multiples of rows)

A1

4

(c) (i) ($k = 1 \Rightarrow$ max reached)
since there are no negative values in top row
provided there are no negative values in top row
" all positive values… " scores E0

E1

(P_{max} =) 36

ft their tableau

B1√ **2**

(ii) k = 3 : new pivot from x-column is $\frac{5}{3}$ used by attempting row operation

ft their pivot if appropriate but must have slack variables

M1

1	0	0	$\frac{41}{5}$	$\frac{3}{5}$	$\frac{9}{5}$	39
0	1	0	$\frac{6}{5}$	$\frac{3}{5}$	$-\frac{1}{5}$	3
0	0	1	$\frac{8}{5}$	$-\frac{1}{5}$	$\frac{2}{5}$	5

first or last row correct
ft one slip from their tableau in part (b) but must use correct pivot

A1√

all correct (condone multiples of rows)

A1 **3**

Optimum reached (or P_{max} = ……)

must have earned M1 and have no negative values in top row

E1

39

ft their tableau

B1√

$x = 3, \quad y = 5, \quad z = 0$
($s = 0$, $t = 0$)

must have correct final tableau

B1 cso **3**

[14]

Q6.

(a) (i) $\frac{4}{-1} - 4;\ \frac{10}{2} = 5;\ \frac{21}{4} = 5\frac{1}{4}$

5 is smallest **positive** ratio

Must see $5 \text{ and } 5\frac{1}{4}$ *plus correct statement*

E1

Pivot = 2

B1

2

(ii)

1	0	$-\frac{1}{2}$	5	0	$\frac{3}{2}$	0	15
0	0	$\frac{3}{2}$	3	1	$\frac{1}{2}$	0	9
0	1	$\frac{1}{2}$	2	0	$\frac{1}{2}$	0	5
0	0	0	−5	0	−2	1	1

row operations (even with wrong pivot)

M1

1st, 2nd or last row correct

A1

another of these correct

A1

all correct (condone multiples of rows)

A1

Negative value in top row
(→ optimum not reached)

must have negative value in their top row

E1

5

(b) (i) New pivot is 'their $\frac{3}{2}$' in *y*-column PI

1	0	0	6	$\frac{1}{3}$	$\frac{5}{3}$	0	18
0	0	1	2	$\frac{2}{3}$	$\frac{1}{3}$	0	6

0	1	0	1	$-\frac{1}{3}$	$\frac{1}{3}$	0	2
0	0	0	−5	0	−2	1	1

or multiple of this

M1

1st, 3rd or 4th row correct

A1

another of these rows correct

A1

all correct (condone multiples of rows)

A1

4

(ii) Optimum value of P reached

must have no negative values in top row

E1

$P = 18$

ft their tableau

B1√

$x = 2$, $y = 6$, $z = 0$

$s = 0$, $t = 0$, $u = 1$

(no more than 2 slips in final tableau for ft)

B1√

$4x + 2y + 3z \le 21$ still has slack

Tableau must indicate u is only slack variable

B1

4

[15]

Q7.

(a) $5x + 3y + 10z \le 15$

$7x + 6y + 4z \le 28$

$4x + 3y + 6z \le 12$

2 inequalities correct

or all 3 LHS & RHS correct but using <

M1

all correct

A1 **2**

(b) (i) Choosing 3 from bottom row as pivot

identified or used

B1

1	6	0	$12 - k$	0	0	2	24
0	1	0	4	1	0	–1	3
0	–1	0	–8	0	1	–2	4
0	$\frac{4}{3}$	1	2	0	0	$\frac{1}{3}$	4

row operations (even with wrong pivot)

M1

one of rows 1, 2, 3 correct

A1

all correct (condone multiples of rows)

A1 **4**

(ii) $12 - k < 0$

their '$12 - k$' < 0

M1

$\Rightarrow k > 12$

SC B1 for $k \geq 13$

A1 **2**

(c) (i)

1	6	0	–8	0	0	2	24
0	1	0	4*	1	0	–1	3
0	–1	0	–8	0	1	–2	4
0	$\frac{4}{3}$	1	2	0	0	$\frac{1}{3}$	4

correct pivot from z column 4 (identified or used)*

M1

1	8	0	0	2	0	0	30
0	$\frac{1}{4}$	0	1	$\frac{1}{4}$	0	$-\frac{1}{4}$	$\frac{3}{4}$
0	1	0	0	2	1	–4	10
0	$\frac{5}{6}$	1	0	$-\frac{1}{2}$	0	$\frac{5}{6}$	$\frac{5}{2}$

one of rows 1, 3 or 4 correct

A1

another of rows 1, 3 or 4 correct

A1

all correct (condone multiples of rows)

A1

4

(ii) Maximum value of P now reached

their tableau must have no negatives in top row

E1

$P = 30, x = 0, y = \frac{5}{2}, z = \frac{3}{4}$

ft their values from their tableau provided at least 2 marks earned in (c)(i)

B1√

$s = 0$, $t = 10$, $u = 0$

condone up to 2 slips in their final tableau

B1cao

3

[15]

Q8.

(a) (i) Slack (variables)

Must be correct word

E1

1

(ii) $2x + 2y + z + s = 14$

Exactly this

B1

1

(b) (i) Pivot from y-column = 1

Identified or seen used by keeping 3rd row fixed

B1

1	–6	0	5	0	4	0	24
0	4	0	–3	1	–2	0	2
0	–1	1	2	0	1	0	6
0	8	0	–5	0	–4	1	5

Row operations, even with wrong pivot

M1

1st, 2nd or 4th row correct

A1

All correct

A1

4

(ii) Still negative value in top row

(only award if this is true for their tableau)

E1

1

(c) (i) Choosing 4 as pivot in x-column

And perhaps dividing by 4 (using their pivot)

M1

1	0	0	$\frac{1}{2}$	$\frac{3}{2}$	1	0	27
0	1	0	$\frac{-3}{4}$	$\frac{1}{4}$	$\frac{-1}{2}$	0	$\frac{1}{2}$
0	0	1	$\frac{5}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{13}{2}$
0	0	0	1	–2	0	1	1

1st, 3rd or 4th row correct ft one slip

A1

1st, 3rd or 4th row (another correct)
ft one slip

A1

All correct (condone multiples of rows)

A1

4

(ii) Optimum now reached (since no negatives in top row)

Or maximum value of P indicated (must have no negatives in top row)

E1

$P = 27$

ft their tableau P

B1ft

$x = \frac{1}{2}$, $y = 6\frac{1}{2}$, $z = 0$

CAO; final tableau "correct" one slip

B1

3

[14]

Q9.

(a)

P	x	y	z	s	t	value
1	–6	–5	–3	0	0	0
0	1	2	k	1	0	8
0	2	10	1	0	1	17

Two slack variables used correctly

M1

1 row correct

A1

all correct

A1

3

May earn in (b)(i)

(b) (i) Pivot in x-column = 1

May be implied by second row unchanged

B1

1	0	7	$6k – 3$	6	0	48
0	1	2	k	1	0	8
0	0	6	$1 – 2k$	–2	1	1

row operations (even with wrong pivot)

M1A1

1st or 3rd row correct all correct

A1

4

(ii) $6k - 3 < 0$

"their" $6k - 3 < 0$

M1

$\Rightarrow k < \frac{1}{2}$

A1

2

(c)

1	0	7	–9	6	0	48
0	1	2	–1	1	0	8
0	0	6	3	–2	1	1

new pivot correct from their tableau and row operations attempted

M1

1	0	25	0	0	3	51
0	1	4	0	$\frac{1}{3}$	$\frac{1}{3}$	$8\frac{1}{3}$
0	0	2	1	$-\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

2 rows correct (may be multiples of rows) usually pivot row & 1 other

A1

all correct (condone multiples of rows)
Condone FT from one slip in (b)(i)

A1

Max P now achieved

Or "optimum", "P_{max} = … "etc"
But must have no negatives in top row

E1

$P = 51$

FT their tableau

B1ft

$x = 8\frac{1}{3},\ y = 0,\ z = \frac{1}{3}$ (all three)

correct values from almost 'correct' tableau (condone one slip)
condone 8.33 or better

B1

3

[15]

Q10.

(a)

P	x	y	z	r	s	Value		
1	−4	5	−6	0	0	0	*B0 if no slack variables used*	B1
0	6	7	–4	1	0	30		B1
0	2	4	–5	0	1	8		B1

3

(b) (i) Both negative when each value is divided by the entry in z-column

E1

1

(ii) Pivot from x-column since value in top row is negative

$\frac{30}{6} = 5, \frac{8}{2} = 4$ and $4 < 5$

Both calculations and comparison needed

E1

Choose 2 as pivot

B1

2

(iii)

1	0	13	−16	0	2	16
0	0	–5	11	1	–3	6
0	1	2	–2½	0	½	4

Row operations keeping pivot row fixed or divided by 2

M1

First or second row correct

A1

All correct
(final row may be 0 2 4 –5 0 1 8)

A1

3

(iv) $x = 4$

B1

$y = 0, z = 0$

B1

2

(v) As z increases, P increases without limit

E1

1

(c) (i) New initial tableau

Q	x	y	z	r	s	Value
1	–4	5	20	0	0	0
0	6	7	–4	1	0	30
0	2	4	–5	0	1	8

B1ft

Revised tableau after one iteration

1 0 13 10 0 2 16

Top row only changed to exactly this

B1

2

(ii) Max $Q = 16$

B1

1

[15]

Q11.

(a) $x + 2y + 3z \le 7$

$2x + y + 4z \le 10$

Exactly this

B1

1

(b) (i) Pivot is 2 in x-column

*Must be ringed or clearly indicated or stated – **not** simply implied*

B1

P	x	y	z	s	t	value
1	0	1	$8 - k$	0	2	20
0	0	1½	1	1	–½	2
0	1	½	2	0	½	5

row operations (even with incorrect pivot) condone one slip

M1

Top or 2nd row correct using correct pivot

A1

All correct (condone multiples of rows)

A1

4

(ii) $8 - k < 0$

Their f(k) < 0

M1

$\Rightarrow k > 8$

SC B1 for $k \geq 9$

A1

2

(c) (i) New pivot from z-column in second row

Stated or possibly implied from tableau

B1ft

P	x	y	z	s	t	value
1	0	4	0	2	1	24
0	0	1½	1	1	–½	2
0	1	–2½	0	–2	1½	1

row operations using "their" ***correct pivot*** *condone 1 slip*

M1

one row (other than pivotal row) correct

A1

all correct (condone multiples of rows)

A1

4

(ii) $P = 24$

Provided no negatives in top row

B1ft

Optimum now reached

Or P_{max} = ...

E1

x = 1, y = 0, z = 2

Only ft if no more than 2 slips in final tableau

B1ft

3

[14]

Q12.

(a) $x + z \leq 9$

$2x + y + 4z \leq 40$

$4x + 2y + 3z \leq 33$

One correct inequality or all using <

M1

All correct

A1

2

(b) (i) Pivot is **1** in z-column

May be implied by use

M1

P	x	y	z	s	t	u	value
1	3	–3	0	5	0	0	45
0	1	0	**1**	1	0	0	9
0	–2	1	0	–4	1	0	4
0	1	2	0	–3	0	1	6

One row correct (other than pivot)

A1

Another row correct (other than pivot)

A1

All correct

A1

4

(ii) (Know optimal value **not** reached) since –3 in <u>top row</u>

E1

1

(c) (i)

1	$4\frac{1}{2}$	0	0	$\frac{1}{2}$	0	$\frac{3}{2}$	54
0	1	0	1	1	0	0	9
0	$-2\frac{1}{2}$	0	0	$-2\frac{1}{2}$	1	$-\frac{1}{2}$	1
0	$\frac{1}{2}$	**1**	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	3

*Next pivot **2** in y-column and perhaps divide by 2*

M1

One row correct (other than pivot)

A1

Another row correct

A1

All correct

A1

4

(ii) Optimum value of P now reached

FT statement if their tableau has negative values in top row

E1ft

$P = 54$, $x = 0$, $y = 3$, $z = 9$

B1ft

$s = 0$, $t = 1$, $u = 0$

All correct and final tableau correct

B1

3

[14]

Q13.

(a) (i) 4 is chosen as pivot

B1

$\frac{20}{4} = 5 < \frac{14}{2} = 7$

and $5 < \frac{8}{1} = 8$

Must have 3 values possibly unsimplified plus comment about smallest (positive) quotient

E1

2

(ii)

P	x	y	z	s	t	u	v	value
1	0	0	0	5	6	0	3	97
0	1	0	0	1	8	0	2	56
0	0	1	0	0	$\frac{3}{4}$	0	$\frac{1}{4}$	5
0	0	0	0	–3	$\frac{1}{2}$	1	$-\frac{1}{2}$	4
0	0	0	1	2	$4\frac{1}{4}$	0	$-\frac{1}{4}$	3

may be left as { 0 0 4 0 0 3 0 1 20 }

B1

or multiples of these rows

B1

SC MI for row operations if wrong pivot used

B1

SC B1 + B1 max ft if pivot row incorrect after ÷ 4

B1

4

(b) Optimum since no negative values in first row

Must have attempted row operations

E1

1

(c) Maximum P = 97

B1ft

x = 56, y = 5, z = 3

B1ft

2

(d) s = 0, t = 0, v = 0, u = 4

B1ft

⇒ only 1 of original inequalities has some slack

Ft if > 1 non-zero slack variables

E1ft

2

[11]

Q14.

(a)

P	x	y	z	s	t	Value		
1	−5	−8	−7	0	0	0	*SCA*	M1
0	3	2	1	1	0	12	*−1 EE*	A2
0	2	4	5	0	1	16		

3

(b) (i) $\frac{12}{2} = 6; \frac{16}{4} = 4$ and $4 < 6$

E1

1

(ii)

1	−1	0	3	0	2	32
0	2	0	−1½	1	−½	4
0	½	1	1¼	0	¼	4

using 4 as pivot and possibly dividing third row by 4

M1

top row correct

A1

second row correct; may have
0 2 4 5 0 1 16

A1

choice of pivot from x-column

pivot = 2 *identified and used*

M1

1	0	0	2¼	½	$1\frac{3}{4}$	34
0	1	0	$-\frac{3}{4}$	½	−¼	2
0	0	1	$1\frac{5}{8}$	−¼	$\frac{3}{8}$	3

row operations

m1

correct or scaled up
0 0 4 6½ –1 1½ 12

A1 **6**

(iii) Max $P = 34$

B1ft

$x = 2, y = 3, z = 0$
all correct

B1 **2**

(iv) Yes - no negative values in first row
no – if negatives in top row

E1ft **1**

[13]

Q15.

(a) $x + 2y \leq 36$
$x + y \leq 20$
$4x + y \leq 39$
One correct, or all inequalities with <

M1

All correct

A1 **2**

(b) (i) Choosing 2 as pivot
And perhaps dividing second row by 2

M1

P	x	y	s	t	u	value
1	$-\frac{1}{2}$	0	$2\frac{1}{2}$	0	0	90
0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	18
0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	2
0	$3\frac{1}{2}$	0	$-\frac{1}{2}$	0	1	21

Row operations

m1

One row correct

A1

All rows correct
(condone multiples of rows)

A1

4

(ii) Negative value in top row $\Rightarrow$ optimum not yet reached

E1

1

(c) (i) New pivot (x – column, 3rd row)
And perhaps multiplying by 2

M1

P	x	y	s	t	u	value
1	0	0	2	1	0	92
0	0	1	1	–1	0	16
0	1	0	–1	2	0	4
0	0	0	3	–7	1	7

Row operations

m1

One row correct

M1

All rows correct

A1

4

(ii) Optimum value reached
(Or not? – if their tableau wrong)

E1

$$\left.\begin{array}{l} P = 92, x = 4, y = 16 \\ s = 0,\ t = 0,\ u = 7 \end{array}\right\}$$

FT 3 values

B1ft

CSO (final tableau must be correct)

B1

3

[14]

Q16.

(a) Introducing slack variables

M1

p x y z r s value

1	–3	–2	–4	0	0	0
0	1	4	**2**	1	0	8
0	2	7	3	0	1	21

–1 EE

A2

3

(b) Choosing correct pivot in z–column

and perhaps dividing by 2

M1

1	–1	6	0	2	0	16
0	$\mathbf{\frac{1}{2}}$	2	1	$\frac{1}{2}$	0	4
	$\frac{1}{2}$	1	0	$-\frac{3}{2}$	1	9

row operations

M1

Correct

A1

3

(c) (i) Need to use x – column for pivot

M1

Choosing correct pivot

A1

1	0	10	2	3	0	24
0	1	4	2	1	0	8
0	0	–1	–1	–2	1	5

row operations

M1

top row

A1

third row

A1

5

(ii) Yes optimal

B1ft

No negative values in top row

E1

2

[13]

Q17.

(a) $3x + 7y \le 33$

$x + 2y \le 10$

$2x + 7y \le 26$

One correct inequality, or all using <

M1

All correct

A1

2

(b) (i) Compare $\frac{33}{3}, \frac{10}{1}, \frac{26}{2}$

E1

Choose smallest positive value $\Rightarrow$ pivot = 1

E1

2

(ii)

P	x	y	r	s	t	Value
1	0	–1	0	4	0	40
0	0	1	1	–3	0	3
0	1	2	0	1	0	10
0	0	$\boxed{3}$	0	–2	1	6

Row operation

M1

Correct one row (other than pivot row)

A1

All correct

A1

next y pivot on $\boxed{3}$

M1

1	0	0	0	$3\frac{1}{3}$	$\frac{1}{3}$	42

0	0	0	1	$-2\frac{1}{3}$	$-\frac{1}{3}$	1
0	1	0	0	$2\frac{1}{3}$	$-\frac{2}{3}$	6
0	0	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	2

Row operation

m1

Correct one row (other than pivot row)

A1

All correct (condone multiples of given rows)
(maximum 6 if y-pivot used first)

A1

7

(iii) No negative number in top row

E1

$P_{max} = 42$

ft if M3 scored and optimum reached

B1ft

$x = 6$ $y = 2$

B1ft

3

[14]

Q18.

(a) $4x + 5y \le 36$

***SCA** at LHS and RHS*

M1

$2x + y \le 12$

All correct with correct inequalities

A1

2

$5x + 2y \le 35$

(b) (i)

P	*x*	*y*	*r*	*s*	*t*	*value*
1	–3	–2	0	0	0	0

0	4	5	1	0	0	36
0	1*	$\frac{1}{2}$	0	$\frac{1}{2}$	0	6
0	5	2	0	0	1	35

Identifying pivot and possibly dividing by 2

M1

1	0	$-\frac{1}{2}$	0	$\frac{3}{2}$	0	18
0	0	3*	1	–2	0	12
0	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	6
0	0	$-\frac{1}{2}$	0	$-\frac{5}{2}$	1	5

Row operations

m1

Correct tableau

A1

Next *y* pivot on 3

M1

1	0	0	$-\frac{1}{6}$	$\frac{7}{6}$	0	20
0	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	0	4
0	1	0	$-\frac{1}{6}$	$\frac{5}{6}$	0	4
0	0	0	$\frac{1}{6}$	$-\frac{17}{6}$	1	7

Row operations

m1

Correct tableau

A1

Optimal since no negative numbers in top row

B1

7

(ii) $P = 20$

B1ft

FT *ONLY if no negs in top row*

B1ft

2

(iii) $x = 4$, $y = 4$
$r = 0$, $s = 0$, $t = 7$ at optimum

B1ft

1

[12]

Q19.

(a)

x	y	s	t	u	P	
2	1	1	0	0	0	20
1	[2]	0	1	0	0	20
5	4	0	0	1	0	60
–2	–3	0	0	0	1	0

M1

A1

2

(b)

3	0	2	–1	0	0	20
1	2	0	1	0	0	20
[3]	0	0	–2	1	0	20
–1	0	0	3	0	2	60

Correct use of their pivot

M1

Row reduction

M1

A1

0	0	2	1	–1	0	0

0	6	0	5	–1	0	40
3	0	0	–2	1	0	20
0	0	0	7	1	6	200

Correct use of their pivot M1

Row reduction M1

A1

$P = \frac{100}{3}$ B1

$x = \frac{20}{3}$, $y = \frac{20}{3}$ B1B1

9

[11]

Q20.

(a)

x	y	z	r	s	t		
3	6	1	1	0	0	0	72
4	2	1	0	1	0	0	48
1	–1	1	0	0	1	0	36
–2	3	–1	0	0	0	1	0

a tableau M1

A1

2

(b) Pivot x, 4 M1

0	18	1	4	–3	0	0	144
4	2	1	0	1	0	0	48
0	–6	3	0	–1	4	0	96
0	8	–1	0	1	0	2	48

row reduction

M1

A1

Pivot z, 3

0	60	0	12	–8	–4	0	336
12	12	0	0	4	–4	0	48
0	–6	3	0	–1	4	0	96
0	18	0	0	2	4	6	240

row reduction

M1

A1

All non-negative in the *P* row
∴ optimal

E1

$P = 40$

B1

$x = 4$, $y = 0$, $z = 32$

B1

$s = t = 0$
$r = 28$

B1

10

[12]

Q21.

(a) (i) All ≥ 0

B1

1

(ii) $P = 2.4$

B1

$n = 0.4$, $y = 0.2$, $z = 0$
All three

B1

2

(b)

x	y	z	r	s	P	
3	4	2	1	0	0	2
1	3•	2	0	1	0	1
–3	–6	–2	0	0	1	0
5•	0	–2	1	–1	0	2
1	3	2	0	1	0	1
–1	0	2	0	2	1	2
5	0	–2	1	–1	0	2
0	15	12	–1	6	0	3
0	0	8	1	9	5	12

Tableau M1A1

Pivot M1

Row reduce M1

All correct A1

Pivot M1

Row reduce M1

All correct A1

All positive
$\therefore\ P = 2.4$
$x = 0.4$, $y = 0.2$, $z = 0$ B1

9

[12]

Q22.

(a)

x	y	z	r	s	P	

$$\left[\begin{array}{cccccc|c} 1 & 3^\circ & 2 & 1 & 0 & 0 & 11 \\ 3 & 4 & 2 & 0 & 1 & 0 & 21 \\ -3 & -6 & -2 & 0 & 0 & 1 & 0 \end{array}\right]$$

Or equivalent

B2,1,0

2

(b)

$$\left[\begin{array}{cccccc|c} 1 & 3^\circ & 2 & 1 & 0 & 0 & 11 \\ 5^\circ & 0 & -2 & -4 & 3 & 0 & 19 \\ -1 & 0 & 2 & 2 & 0 & 1 & 22 \end{array}\right]$$

Row reduction

M1

Pivot

M1

All correct

A1

$$\left[\begin{array}{cccccc|c} 0 & 15 & 12 & 9 & -3 & 0 & 36 \\ 5 & 0 & -2 & -4 & 3 & 0 & 19 \\ 0 & 0 & 8 & 6 & 3 & 5 & 129 \end{array}\right]$$

Row reduction

M1

Pivot

M1

All correct

A1

$P = \frac{129}{5} = \underline{25.8}$

B1

$z = 0$, $x = \frac{19}{5}$, $y = \frac{12}{5}$

B1

8

<u>Alternative answer</u>

(a)

P	x	y	z	r	s	
1	–3	–6	–2	0	0	0
0	1	3°	2	1	0	11
0	3	4	2	0	1	21

Or equivalent

(B2,1,0)

(b)

1	–1	0	2	2	0	22
0	1	3	2	1	0	11
0	5	0	–2	–4	3	19

Row reduction

(M1)

Pivot

(M1)

All correct

(A1)

5	0	0	8	6	3	129
0	0	15	12	9	–3	36
0	5	0	–2	–4	3	19

Row reduction

(M1)

Pivot

(M1)

All correct

(A1)

$P = \frac{129}{5} = 25.8$

(B1)

$z = 0$ $x = \frac{19}{5}$,$y = \frac{12}{5}$

Or equivalent

(B1)

[10]

Q23.

(a) I > III dominance

E1

1

(b) $v \le 4p_1 + 2p_2$

Ignore any p_3 values

B1 × 3

3

$v \le 2p_1 + 10p_2$

$v \le 6p_1 + p_2$

(c) (i)

p	v	p_1	p_2	r	s	t	u	
1	0	–2	–10	0	0	1	0	0
0	0	1	1	1	0	0	0	1
0	0	–2	8	0	1	–1	0	0
0	1	–2	–10	0	0	1	0	0
0	0	–4	9	0	0	–1	1	0

For row reduction

M1

For correctly having only 1 non-zero in v

A1

Top row correct

A1*

All correct
(*)OE

A1*

4

(ii) Negative values

E1

1

(iii) $5P = 18 \quad P = 3\frac{3}{5}$

B1

$p_1 = \frac{4}{5}$

B1

$p_2 = \frac{1}{5}$

B1

3

(d) B plays

I	prob	p
II	prob	q
III	prob	$(1 - p - q)$

M1

$\therefore \; 4p + 2q + 6(1 - p - q) = 3\frac{3}{5}$

$2p + 10q + 1 - p - q = 3\frac{3}{5}$

M1

$\Rightarrow 5p + 10q = 6$

$\underline{5p + 45q = 13}$

A1

$\Rightarrow q = \frac{1}{5}$

B1

$p = \frac{4}{5}$

B1

6

[18]

Q24.

(a) $x \quad y \quad r \quad s \quad P$

2•	1	1	0	0	32
1	1	0	1	0	20
–8	–7	0	0	1	20

M1

All correct

A1

2

(b) 2 1 1 0 0 32

Row reduction

M1

0 1• –1 2 0 8

Using pivot

M1

0 –3 4 0 1 128

2 0 2 –2 0 24

A1

Row reduction

M1

0 1 –1 2 0 8

Using pivot

M1

0 0 1 6 1 152

$P = 152$

B1

$\left.\begin{array}{l} x = 12 \\ y = 8 \end{array}\right\}$

B1

7

[9]

Q25.

(a) $4x + 3y \leq 33$

M1

$-x + y \leq 4$

$2x + 5y \leq 27$

A1

2

(b)

4*	3	1	0	0	0	33	(R_1)
–1	1	0	1	0	0	4	(R_2)
2	5	0	0	1	0	27	(R_3)
–2	–2	0	0	0	1	0	

Pivot about *x*, 4

4	3	1	0	0	0	33	R_1
0	7	1	4	0	0	49	$4R_2 + R_1$
0	7*	–1	0	2	0	21	$2R_3 – R_1$
0	–1	1	0	0	2	33	$2R_4 + R_1$

M1

m1

A1

Pivot about *x*, 7

28	0	10	0	–6	0	168	$7R_1 – 3R_3$
0	0	2	4	–2	0	28	$R_2 – R_3$
0	7	–1	0	2	0	21	R_3
0	0	6	0	2	14	252	$7R_4 + R_3$

M1

m1

A1

All positive, therefore optimal

Alternative

P	*x*	*y*	*r*	*s*	*t*	*u*	
1	–2	–2	0	0	0	0	
0	4	3	1	0	0	33	M1
0	–1	1	0	1	0	4	
0	2	5	0	0	1	27	
1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{33}{2}$	M1
0	1	$\frac{3}{4}$	$\frac{1}{4}$	0	0	$\frac{33}{4}$	A1
0	0	$\frac{7}{4}$	$\frac{1}{4}$	1	0	$\frac{49}{4}$	
0	0	$\frac{7}{2}$	$-\frac{1}{2}$	0	1	$\frac{21}{2}$	M1
1	0	0	$\frac{3}{7}$	0	$\frac{1}{7}$	18	m1
0	1	0	$\frac{5}{14}$	0	$-\frac{3}{14}$	6	
0	0	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	7	A1
0	0	1	$-\frac{1}{7}$	0	$\frac{2}{7}$	3	B1

B1

7

(c) $14P = 252$

$P = 18$

B1

$y = 3$, $x = 6$

for both

B1

2

[11]

Q26.

(a)

x	y	z	r	s	P	1
8	5	2	1	0	0	3
4	6*	9	0	1	0	2
−4	−5	−3	0	0	1	0

B1 B1

2

(b) Pivot y, 6

R_2	4	6	9	0	1	0	2
$6R_1 - 5R_2$	28*	0	−33	6	−5	0	8
$6R_3 + 5R_2$	−4	0	27	0	5	6	10

Choosing pivot

M1

Attempt zeroes

M1

CAO

A1

Pivot x, 28

$7R_1 - R_2$	0	42	96	−6	12	0	6
R_2	28	0	−33	6	−5	0	8
$7R_3 + R_2$	0	0	156	6	30	42	78

As above

M1 M1 M1

CAO

A1

All +ve ∴ optimal

M1

$P = \frac{78}{42} = \frac{13}{7}$

If all + ve

A1F

$z = 0$

$x = \frac{8}{28} = \frac{2}{7}$

$y = \frac{6}{42} = \frac{1}{7}$

As fractions

A1F

9

[11]

Q27.

(a) Initial tableau

x	y	z	r	s	P	
3	2*	1	1	0	0	24
2	1	2	0	1	0	20
−3	−5	−4	0	0	1	0

B1

B1

2

(b) Pivot about y, 2

choose pivot

M1

	x	y	z	r	s	P	
	3	2	1	1	0	0	24
$2R_2 - R_1$	1	0	3	−1	2	0	16
$2R_3 + 5R_1$	9	0	−3	5	0	2	120

make zeros
o.e./c.a.o

m1}

A1}

Pivot about z, 3

choose pivot

M1

	x	y	z	r	s	P	
$3R_1 - R_2$	8	6	0	4	–2	0	56
	1	0	3	–1	2	0	16
$R_3 + R_2$	10	0	0	4	2	2	136

make zeros
o.e./c.a.o

m1}

A1}

All positive, ∴ optimal

M1

$P = \frac{136}{2} = 68, \quad x = 0$

if all positive

A1F

$y = \frac{56}{6} = \frac{28}{3}$ }

$z = \frac{16}{3}$ }

both as fractions

A1F

9

Alternative

(a)

P	x	y	z	s	t	
1	–3	–5	–4	0	0	0
0	3	2	1	1	0	24
0	2	1	2	0	1	20

B1

B2

2

(b)

P	x	y	z	s	t	
1	$\frac{9}{2}$	0	$-\frac{3}{2}$	$\frac{5}{2}$	0	60
0	$\frac{3}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	12
0	$\frac{1}{2}$	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	8
1	5	0	0	2	1	68
0	$\frac{4}{3}$	1	0	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{28}{3}$
0	$\frac{1}{3}$	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	$\frac{16}{3}$

choose pivot

M1

make zeros

m1

c.a.o

A1

Top row all positive, ∴ optimal

M1

∴ $P = 68$, $x = 0$

if all positive

A1F

$y = \frac{28}{3}$ }

$z = \frac{16}{3}$

both as fractions

A1F

9

[11]

Q28.

(a) Slack variables

B1

1

(b) 4

B1

1

(c) ***P x y z s t u v***

1	0	0	0	1	0	1	2		100
0	0	0	1	0	0	1	0		55
0	0	0	0	2½	1	–½	1½		5
0	**1**	0	0	–½	0	½	½		20
0	0	1	0	½	0	–½	3½		10

Choice of pivot and making it 1

M1 A1

Row operations

M1 A1

A1

5

(d) Optimal since no negatives in top row

B1

$P = 100$

Ft

B1 ft

$x = 20$, $y = 10$, $z = 55$

ft

B1 ft

3

(e) $s = u = v = 0$, $t = 5$

B1

Slack in one inequality

B1

2

[12]

Q29.

(a) Maximise $P = 15x + 10y + 10z$

B1

Subject to

($x \geq 0$ $y \geq 0$ $z \geq 0$)

$x + 2y + z \leq 60$

one inequality

B1

$x + y + z \leq 55$

$3x + 6y + 2z \leq 140$

the two others

B1 **3**

(b)

1	–15	–10	–10	0	0	0	0
0	1	2	1	1	0	0	60
0	1	1	1	0	1	0	55
0	3	6	2	0	0	1	140

coefficients of x, y, z

B1

appropriate slacks

B1 **2**

(c)

1	–5	0	0	0	10	0	55 0
0	0	1	0	1	–1	0	5
0	1	1	**1**	0	1	0	55
0	1	4	0	0	–2	1	30

Pivot

M1A1

row operations

M1A1

A1 **5**

(d)

1	0	20	0	0	0	5	700
0	0	1	0	1	–1	0	5
0	0	–3	1	0	3	–1	25
0	**1**	4	0	0	–2	1	30

M1A1 **4**

(e) Maximum profit £700

Ft

B1ft

Make 30 Xtrafast, 0 Yourelax, 25 Zizz

B1ft

ft

e.g. Not helpful for people who want Yourelax.

B1

3

[17]

Q30.

(a) **Maximise $P = 20x + 10y + 30z$**

M1

Subject to

$x \geq 0$, $y \geq 0$, $z \geq 0$, $2x + y + 2z \leq 110$

A1

$x + y + z \leq 60$, $2x + 3y + 3z \leq 140$

A1

3

(b)

P	x	y	z	s	t	u	
1	−20	−10	−30	0	0	0	0
0	2	1	2	1	0	0	110
0	1	1	1	0	1	0	60
0	2	3	3	0	0	1	140

M1 A1

2

(c)

1	0	0	−10	10	0	0	1100
0	1	½	1	½	0	0	55
0	0	½	0	−½	1	0	5
0	0	2	1	−1	0	1	30

Pivot → 1

M1 A1

Subtracting rows

M1

A1

A1

5

(d)

1	0	20	0	0	0	10	1400
0	1	–1½	0	1½	0	–1	25
0	0	½	0	–½	1	0	5
0	0	2	1	–1	0	1	30

M1

A1

ft

A1

3

(e) Make 25 stools, 0 armchairs and 30 settees. Impractical because people want matching armchairs

M1ft A1ft B1

3

[16]

Q31.

(a) $P = 3x + 2y + 2z$

B1

1

(b) (i)

P	x	y	z	s	t	u	
1	0	1	–½	0	1½	0	225
0	0	0	½	1	–½	0	5
0	1	1	½	0	½	0	75
0	0	1	2	0	–1	1	30

Choice of pivot and pivot → 1
Row deductions

M1

A1

M1

A1

A1

5

(ii) Still a negative in top row

B1 1

(c)

P	x	y	z	s	t	u	
1	*0*	*1*	*0*	*1*	*1*	*0*	*230*
0	*0*	*0*	1	2	–1	0	10
0	1	1	0	–1	1	0	70
0	0	1	0	–4	1	1	10

M1

A1

A1

A1 4

(d) Maximum of P is 230 at (70,0,10)

B1ft

ft near misses

B1ft 2

(e) Slack variable $u \neq 0$.

(or test each inequality)

M1

Third inequality has slack;
i.e. $2x + 3y + 3z \leq 180$

A1 2

[15]

Q32.

(a) (i)

P	x	y	s	t	u	
1	0	0	0	1¼	–¼	18
0	0	0	1	½	½	14
0	0	1	0	¼	–¼	3
0	1	0	0	¾	¼	12

Choice of pivot

B1 5

Pivot → 1

B1

Row manipulations

M1

A1

A1

(ii) Still a negative in top row

B1

1

(b) (i) P's maximum is 25

B1

Reached at (5,10)

B1

2

(ii) T_0: O

Any two

B1

T_1: D

Another

B1

T_2: C

Another

B1

3

T_3: B

(iii) Start by pivoting in y's column

M1

A1

2

[13]

Q33.

(a) Pivot around the 4 in z's column to give

Choice of pivot and reducing to 1 in pivot position

M1 A1

P x y z s t u v

Using pivot to abtain 0s elsewhere in column

M1

1	0	0	0	1	2	1	0	79
0	1	0	0	0	0	1	0	53
0	0	1	0	$\frac{1}{2}$	$1\frac{1}{2}$	$-\frac{1}{2}$	0	8
0	0	0	1	$-\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	9
0	0	0	0	$1\frac{1}{4}$	$1\frac{3}{4}$	$\frac{3}{4}$	1	1

A1

A1

5

(b) Maximum because no top-row entries are negative

B1

Maximum value is 79

B1ft

reached when $x = 53$, $y = 8$ and $z = 9$

B1

3

(c) 4 inequalities

B1

Only $v \neq 0$ and so there is slack in just one

B1

2

[10]

Q34.

(a) $x + 2y \leq 20$

M1

$x + y \leq 11$

A1

$4x + y \leq 23$

$P = 2x + 3y$

B1

3

(b) Pivot at 2 to give

M1

P	x	y	s	t	u	
1	$-\frac{1}{2}$	0	$+1\frac{1}{2}$	0	0	30
0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	10
0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	1
0	$3\frac{1}{2}$	0	$-\frac{1}{2}$	0	1	13

pivot → 1
entries → 0
cao

M1 M1 A1

Point $x = 0$ $y = 10$

A1ft

Still negative in top row

B1

6

(c) Pivot to give

P	x	y	s	t	u	
1	0	0	1	1	0	31
0	0	1	1	–1	0	9
0	1	0	–1	2	0	2
0	0	0	3	–7	1	6

M1 A1 A1ft

max of 31 at (2, 9)

B1

$s = 0$, $t = 0$, $u = 6$

B1

5

[14]

Q35.

(a) $x + 12y + 4z \leq 100$,

M1

$x + 2y + z \leq 40$

A1

$3y + 2z \leq 30$

A1

3

(b) $P = 10x + 40y + 40z$

B1

1

(c) (i) $s = 60$, $t = 0$, $u = 30$

M1 A1

2

(ii) Negative coefficients in first row

B1

1

(iii)

P	x	y	z	s	t	u	
1	0	25	0	0	1	15	850
0	0	5.5	0	1	–1	–1.5	15
0	1	0.5	0	0	1	–0.5	25
0	0	1.5	1	0	0	0.5	15

Pivot
Rows

M1 M1 A1 A1

4

(iv) P takes maximum £850

B1

at (25, 0, 15)

B1 B1

3

[14]

Q36.

(a) P = $9x + 10y$

$x + y + s = 20$

M1

$3x + 2y + t = 48$

A1

(b) Bring in x

$P = 9(16 - \frac{2}{3}y - \frac{1}{3}t) + 10y$

M1/A2

$= 144 + 4y - 3t$

$\frac{1}{3}y + s - \frac{1}{3}t = 4$

$x + \frac{2}{3}y + \frac{1}{3}t = 16$

(c) (i) negative coefficients of s and x in objective

B1

(ii) Solution: $P = 200$ at (0,20)

B1B1

(iii) Only 1 iteration would have been needed

B1

9

[9]

Q37.

(a) $4x + 5y + s = 20$

M1

$3x + 2y + t = 12$

A1

2

(b) Bring in x

$P = (4 - \frac{2}{3}y - \frac{1}{3}t) + y$

for determining the pivot

M1

$= 4 + \frac{1}{3}y - \frac{1}{3}t$

A1

$\frac{7}{3}y + s - \frac{4}{3}t = 4$

A1

$x + \frac{2}{3}y + \frac{1}{3}t = 4$

A1

4

(c) $P = 4$ at (4, 0)

B1f.t.

$s = 4$, showing slack in first constraint.

B1 4; 0f.t.

$t = 0$, showing second constraint tight.

B1 correct f.t.

3

[9]

Q38.

(a) (b) (c) (The solution need not be in tabular form)

P	x	y	s_1	s_2	RHS
1	–300	–200	0	0	0
0	3	1	1	0	3000
0	6	5	0	1	12000

slacks

B2

1	0	–100	100	0	300000
0	1	1/3	1/3	0	1000
0	0	3	–2	1	6000

choice of pivot element

M1

pivot (–1 each error)

A2

1	0	0	33.33	33.33	500000
0	1	0	5/9	–1/9	333.33

0	0	1	–2/3	1/3	2000

choice of pivot element

M1

pivot (–1 each error)

A2

OR

1	–60	0	0	40	480000
0	1.8	0	1	–0.2	600
0	1.2	1	0	0.2	2400

1	0	0	33.33	33.33	500000
0	1	0	5/9	–1/9	333.33
0	0	1	–2/3	1/3	2000

Solution: $x = 333\frac{1}{3}$, $y = 2000$

Answer

M1

clearly stated

A1

10

(d) resource 1 unused = 3000; 0; 0

or 3000; 600 0

resource 2 unused = 12000; 6000 0

or 12000; 0 0

initial values not

B1

needed explicitly

B1

2

[12]

Q39.

(a) P = 9x + l0y

$x + y + s = 20$

M1

$4x + 5y + t = 94$

slack variables

A1

2

(b) Bringing in y

$P = 9x + 10(18.8 - 0.8x - 0.2t)$
$= 188 + x - 2t$
$0.2x + s - 0.2t = 1.2$
$0.8x + y + 0.2t = 18.8$

first pivot

M1

(– 1 each error)

A2

Bringing in x

$P = 188 + (6 - 5s + t) - 2t$
$= 194 - 5s - t$
$x + 5s - t = 6$
$y - 4s + t = 14$

second pivot

M1

(– 1 each error)

A2

Solution: $P = 194$ at (6, 14)

B1 B1

8

or (one mark off)

Bring in x

$P = 9(20 - y - s) + 10y$
$= 180 + y - 9s$
$x + y + s = 20$
$y - 4s + t = 14$

Bring in y

$P = 180 + (14 + 4s - T) - 9s$
$= 194 - 5s - t$
$x + 5s - t = 6$
$y - 4s + t = 14$

[10]

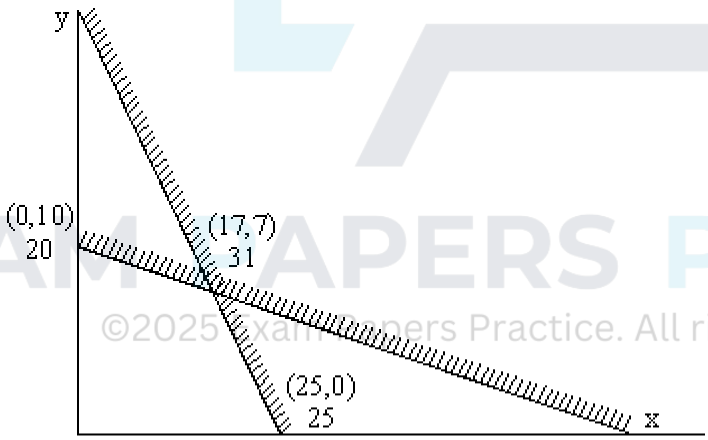
In tableau form:

P	x	y	s	t	RHS
1	–9	–10	0	0	0
0	1	1	1	0	20
0	4	5	0	1	94
1	–1	0	0	2	188
0	0.2	0	1	–0.2	1.2
0	0.8	1	0	0.2	18.8
1	0	0	5	1	194
0	1	0	5	–1	6
0	0	1	–4	1	14

P	x	y	s	t	RHS
1	–9	–10	0	0	0
0	1	1	1	0	20
0	4	5	0	1	94
1	0	–1	9	0	180
0	1	1	1	0	20
0	0	1	–4	1	14
1	0	0	5	1	194
0	1	0	5	–1	6
0	0	1	–4	1	14

Q40.

(a)



scales and labels B1

Lines B1 B1

Shading B1

Solution B1

(b) $P - x - 2y = 0$ $\qquad$ $P - \frac{11}{17}x + \frac{2}{17}s_1 = 20$

Sca

M1

pivoting on 17y

A1

$3x + 17y + s_1 = 170$ $\frac{3}{17}x + y + \frac{1}{17}s_1 = 10$

(−1 each error)

A2

(0,10)

B1

$7x + 8y + s_2 = 175$ $\frac{95}{17}x - \frac{8}{17}s_1 + s_2 = 95$

or $P - \frac{8}{7}y + \frac{1}{7}s_2 = 25$

sca

M1

pivoting 7x

A1

$\frac{95}{7}y + s_1 - \frac{3}{7}s_2 = 95$

(−1 each error)

A2

(25,0)

B1

(c) both = 0

B1

[11]

Q41.

(a) Max 80c + 350t

B1

st 20c + 100t ≤ 10000

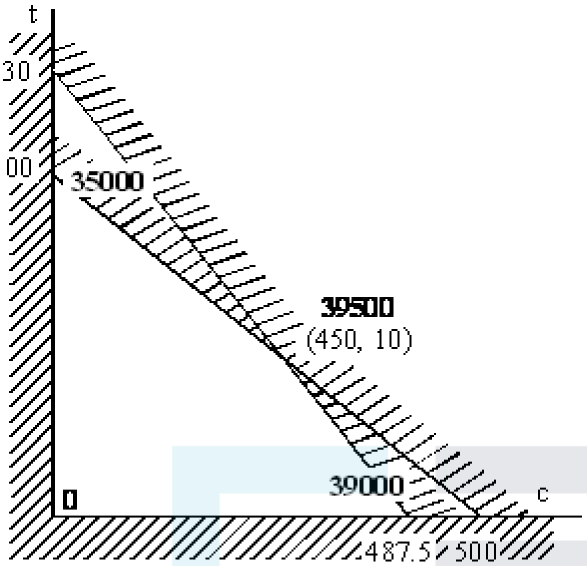
B1

4c+15t ≤ 1950

B1

c, t ≥ 0

(b)



axes labelled and scaled

B1

finance line

B1

hours line

B1

shading

B1

method for optimising
(as indicated or by income lines)

M1

450 chairs and 10 tables

A1

(c) unused finance and unused hours both zero

B1 both

B1

(d) too many chairs per table
(or reasonable alternatives, eg. Lack of availability of tables)

B1

[12]

Examiner reports

Q2.

A number of students failed to score the marks in part (a) as they simply converted the inequalities into equations. Despite reference to the requirements for question (b)(i) in previous reports on the exam for this unit, many students failed to give a fully correct answer. Part (b)(ii) was generally well done. For part (c)(i), students should justify their choice of pivot as a matter of course in any Simplex question, and many students failed to either justify or state the pivot that they were using. Although many students arrived at a correct final tableau in part (c)(ii), a significant number failed to state that an optimum solution had been obtained, and also failed to give the values of x, y and z.

Q3.

(a) A number of students failed to score the marks in this part as they simply converted the inequalities into equations. (b) Despite reference to the requirements for this question in previous examiners reports, many students failed to justify the pivot chosen. (c) Again, many students failed to justify their choice of pivot. This should be done as a matter of course in any Simplex question, and many students failed to either justify or state the pivot that they were using. (ii) Although many students arrived at a correct final tableau, a significant number failed to state that an optimum solution had been obtained and, also, failed to give the values of x, y and z.

Q4.

Part (a)(i) The column and pivot for the next iteration were usually identified correctly but the explanation was often inadequate for full marks. Some indication had to be made as to why the negative quotient had been rejected; the calculations 2 / 2 and 3 / 6 needed to be shown with a statement that 0.5 was the smallest **positive** ratio. Some who did this calculation omitted to say that 6 was the pivot; simply drawing an arrow pointing to the corresponding row was insufficient.

Part (a)(ii) The row operations were usually carried out accurately and most candidates obtained the correct tableau for the next iteration.

Part (b)(i) Most made a suitable comment about there being no negative values in the objective row and therefore the optimal value had not yet been reached; it was incorrect to say that “the values in the top row were all positive”.

Part (b)(ii) The number of incorrect answers suggested that many were “guessing” the number of inequalities that still had slack.

Part (c)(i) Finding the value of P, even from an incorrect tableau, earned most candidates a mark. The values of x, y and z needed to be clearly stated but some neglected to include $y = 0$, even when their previous tableau was correct.

Part (c)(ii) The values for P, x, y and z from part (c)(i) needed to be substituted into the given expression in order to find the value of k. Consequently most scored a method mark here even if their values were incorrect.

Q5.

The printed grids seemed to improve the overall presentation of the Simplex method this year and, because students generally took more care, the overall accuracy seemed much better, particularly in the first tableau.

(a) Some students omitted slack variables by writing all 0s in the s and t columns.

(b) The pivot from the bottom row of the y-column was usually identified correctly, but careless arithmetic prevented a few from scoring full marks.

(c) (i) Many students seemed so concerned with the explanation as to why the maximum had been achieved that they forgot to write down the maximum value of P. Many students seemed confused between the terms "positive" and "non-negative" and lost the mark for the explanation.

(ii) Quite a few students selected the wrong value as their new pivot, and scored no further marks in the question. Those students who found the next pivot correctly often made at least one arithmetic slip and so only the strongest students completed the second iteration correctly. Part of the expected interpretation was to stat that the maximum value of P had now been reached, but very few students realised this. The values of P, x, y and z needed to be stated but some neglected to state that $z = 0$, even when their previous tableau was correct.

Q6.

In part (a)(i) the pivot was usually identified correctly but the explanation was often inadequate for full marks. Some indication needed to be made as to why the negative quotient had been rejected; the calculations 10/2 and 21/4 needed to be shown with a statement that 5 was the smallest positive ratio.

In (a)(ii) the row operations were usually carried out accurately and most candidates made a suitable comment about the negative value in the objective row implying that the optimal value had not yet been reached.

In part (b)(i) those candidates who found the next correct pivot were usually successful in performing a correct second iteration, although some arithmetical slips did occur.

In (b)(ii) finding the value of P earned most candidates a mark, even from an incorrect tableau. Part of the interpretation was to say that the maximum value of P had now been reached, provided there were still no negative values in the top row of their matrix. The values of x, y and z needed to be stated but some neglected to state that $z = 0$, even when their previous tableau was correct. Several candidates wrote down the value of the slack variable, u, instead of writing down the actual inequality that still had slack.

Q7.

(a) Some candidates wrote down incorrect inequalities and quite a large number wrote down equations involving the slack variables s, t and u.

(b) (i) The pivot from the bottom row was usually identified correctly but careless arithmetic prevented many from scoring full marks.

(ii) Although many wrote down that $12 - k$ was negative, it was common to see errors in handling the inequality, with quite a number not concluding that $k > 12$.

(c) (i) A large number of candidates selected –8 as their new pivot, and scored no further marks in the question. Those candidates who found the next pivot correctly often made some arithmetical slips and so only the very best candidates completed the second iteration correctly.

(ii) Part of the interpretation was to say that the maximum value of P had now been reached, but very few candidates realised this. The values of P, x, y and z needed to be stated but some neglected to state that $x = 0$, even when their previous tableau was correct. Quite a number of candidates failed to find the correct values of the slack variables.

Q8.

In part (a)(i), most candidates were familiar with the term "slack variables" but quite a few could not remember the correct word. In part (a)(ii), quite a few wrote down an inequality rather than an equation involving the given variables.

It was encouraging to see that most candidates were able to find the correct pivot in part (b)(i) and many were able to perform the first iteration without many errors. This was possibly due to the fact that the calculations only involved integers. In part (b)(ii), sometimes a statement such as "there are negative values" was given and scored no marks unless the first row, or objective row, was mentioned and actually contained a negative entry.

Those candidates who chose the correct pivot were usually successful in performing a second iteration in part (c)(i), although some arithmetical slips occurred. Part of the interpretation of the final tableau in part (c)(ii) should have included the fact that the optimum or maximum value of P had now been achieved. Credit was given for stating the value of P for their tableau, but only those with a correct final tableau were able to score the mark for the correct values of x, y and z which maximised P.

Q9.

Part (a) Most candidates formed the correct initial tableau, although some failed to include any slack variables and scored no marks. Some misunderstood the request in this part of the question and wrote down three equations introducing the slack variables s and t. Full credit was given if they produced the initial Simplex tableau as their first step in part (b).

Part (b)(i)Those who chose the incorrect pivot could make little progress. However, most candidates chose the correct pivot and answered this part of the question well, apart from a few who made numerical slips.

Part (b)(ii) Many more candidates than on a previous occasion, when a similar question was set, realised that the entry in the top row had to be negative; they formed and solved the correct inequality writing $6k - 3 < 0$ and hence that $k < 0.5$.

Part (c) Those with the correct tableaux, or who made an arithmetic slip in one of their rows, were able to score full marks for finding the values of P and the variables x, y and z. Some failed to state that $y = 0$ and others omitted the value of z. Part of the interpretation of the final tableau was a statement that the optimum had now been reached and many candidates failed to score the mark for this.

Q10.

In part (a) most candidates were able to form the initial tableau, but a few neglected to introduce slack variables and scored no marks. Others had a negative value in the P column and then usually had an incorrect first row.

Explanations in part (b)(i) were often poor. It was insufficient to say that “the values in the z-column were all negative”. In part (b)(ii) some comparison of 30/6 and 8/2 was expected with the smaller of these quotients determining that the pivot was 2. Explanations were often poor and far too many felt it was sufficient to put a circle around the pivot with no explanation. In part (b)(iii) those candidates who had the correct pivot were usually successful in performing a correct iteration, although some arithmetical slips occurred. The value of x in part (b)(iv) was usually correct but often the incorrect values of y and z were given. In part (b)(v) most candidates were happy to make some statement about the entries in the objective row being non-negative, but they had not answered the question. It was necessary to realise that z could increase without limit and consequently so could P.

The final part of this question, part (c), was a good source of marks for most candidates, but it was surprising how many gave answers such as –16 for the maximum value of Q.

Q11.

Several candidates had the wrong inequality signs in part (a) and others included s and t in their answers.

Many candidates failed to indicate the pivot in part (b)(i) and lost a mark; many having found the quotients 7/1 and 10/2 gave the pivot as 5; others simply drew an arrow pointing horizontally towards the 10, which was insufficient. Those who chose the incorrect pivot could make little progress. However, apart from a few who made numerical slips, most candidates answered this part of the question well. In part (b)(ii), most were unable to see why $8 - k < 0$ and hence that $k > 8$.

In part (c), those with the correct tableaux, or who made an arithmetic slip in one of their rows, were able to score full marks for finding the values of P and the variables x, y and z. Some failed to state that $y = 0$ and others omitted the value of z. Part of the interpretation of the final tableau was a statement that the optimum had now been reached.

Q12.

Part (a) Most candidates were able to write down the correct inequalities, although some introduced the slack variables and others wrote equations or used “less than” rather than “less than or equal to”.

Part (b) The pivot was usually found correctly for the first iteration, although some used a value from the x-column, despite the clear instruction. Since this had value 1, the row operations were fairly easy and most candidates were successful in performing the first iteration.

Part (c) Those candidates who reduced the second pivot to 1 and who worked with fractions were usually more successful than those who chose not to use elementary row operations. Some caused problems for themselves by forming negative multiples of the rows. It was then common to see the incorrect pivot being used or the incorrect conclusion being made regarding optimality. Part of the interpretation was to state that the maximum value of P had been reached, but most candidates failed to make such a statement. Others found the values of the slack variables but failed to give the values for x, y and z when interpreting the final tableau.

Q13.

In part (a)(i), candidates made vague statements such as "it has the smallest value when divided into the value" when explaining why 4 was the pivot. It was expected that at least three complete divisions would be performed with a comment about the smallest positive quotient being selected and hence 4 being the pivot.

In part (a)(ii), those who chose the incorrect pivot could make little progress. However, apart from a few who made numerical slips, most candidates answered this part of the question well.

In part (b), Most were able to explain why the optimum had been reached.

In part (c), those with the correct tableau, or those who made an arithmetic slip in one of their rows, were able to score full marks for finding the value of P and the other variables.

In part (d), many failed to give the value of $v = 4$, and a large number of candidates seemed unaware of how the values of the slack variables related to the original inequalities.

Q14.

In part (a), most candidates were able to form the initial tableau, but a few neglected to introduce slack variables and scored no marks. Explanations in part (b)(i) were often poor. Some comparison of 16/4 with 12/2 was expected, with the smaller of these quotients determining that the pivot was 4. The mark scheme for part (b)(ii) gives an indication of the preferred layout of the tableaux. Those candidates who reduced the pivot to 1 and who worked with fractions were usually more successful than those who chose not to use elementary row operations. In part (b)(iii), most candidates were able to find the value of P at the maximum point. Several omitted to say that $z = 0$, and, although many realised that the entries in the objective row were non-negative, they sometimes failed to state that this implied the maximum had been reached and therefore had not answered the question.

Q15.

In part (a), some candidates did not have the correct inequality sign and others included the slack variables in either an equation or an inequality.

A small number of candidates seemed unprepared for the Simplex method in part (b).

However, apart from a few who made numerical slips, most candidates answered this part of the question well. Most were able to explain why the optimum had not yet been reached.

In part (c)(i), quite a few candidates failed to identify the correct pivot at the second iteration stage and scored no marks for their row operations. It is good practice to state the value of the pivot at each iteration in order to convince the examiner that the correct method is being used. Those candidates who used unorthodox tableaux usually fared worst since they often made more errors in their calculations. When asked to interpret the final tableau in part (c)(iii), most candidates failed to say that P had reached a maximum or optimum value; others neglected to give the value of each of the slack variables. Consequently very few indeed scored full marks in this part of the question.

Q16.

Practically everyone seemed familiar with the Simplex method. Apart from a few numerical slips, most candidates scored high marks on this question. A few weaker candidates did not seem to realise how a pivot is selected from a given column and all candidates need to be encouraged to state the value of the pivot at each stage.

Q17.

A small number of candidates seemed unprepared for the Simplex method. However, apart from a few who made numerical slips, most candidates scored high marks on this question. A few candidates did not seem to realise how a pivot is selected from a given column and all candidates need to be encouraged to state the value of the pivot at each stage.

Q18.

Part (a) The inequalities were often given as “equations” usually involving the slack variables. Quite a few wrote <, > or ≥ instead of ≤.

Part (b) The Simplex method seemed to be well understood and, apart from a few arithmetic slips, the iterations were done correctly in part (i). The most successful were those who used fractions and elementary row operations. Some evidently have no idea of how the pivot can be identified, but these are clearly in the minority. Since the tableau in the question is the form which is likely to be given in future questions, it would be good for candidates to practise working with a grid in this form. Full credit was given for finding values of the variables in parts (ii) and (iii) provided that no negative values appeared in the top row of the final tableau.

Q19.

A surprising number of candidates failed to score the first two marks in this question because they introduced the slack variables correctly but did not display the information in a Simplex tableau as instructed. In part (b) there was a variety of orders of choice of pivots though all their choices were capable of leading to the correct solution. Candidates should be encouraged to use the variable with the largest negative coefficient as the pivoting variable as this will lead to the correct solution of the algorithm in the minimum number of passes. Candidates should check to see if their solution has an implication of a negative slack variable as a further confirmation to see if their solution is optimum.

Q20.

There was a surprising number of candidates who failed to score the first two marks in this question by merely adding the slack variables but not displaying the revised equations in a tableau. There was a variety in the order of choice of pivots, all of which could lead to the correct solution. Candidates should be encouraged to use the greatest negative variable as the pivoting variable as this will lead to the correct solution in the minimum number of passes of the algorithm. Many candidates did not state the values of the slack variables despite this being a specific request in the question. Candidates should also make it clear that they know that an optimum solution has been reached by referring to non-negativity in the profit row.

Q21.

This question was well answered. Candidates were able to demonstrate that they knew the Simplex algorithm and were able to apply it properly. In particular, most candidates chose the correct pivot at each stage. Even weaker candidates were able to make a good attempt at this question.

Q22.

A surprising number of candidates failed to score the first two marks in this question by merely putting in slack variables but not displaying the revised equations in a tableau. There were several ways of choosing the pivots so as to arrive at the correct solution. Candidates who chose to use the largest negative variable as the pivoting variable benefited, as this leads to the correct solution in the minimum number of passes of the algorithm. Candidates should have checked if their solution had a negative coefficient for a slack variable; this happened when the z column was used for the first choice of pivot.

Q23.

This question combined Game Theory and the Simplex algorithm. Parts (a) and (b) were well answered. Candidates knew the method for part (c)(i) but there were many numerical slips. Although strict use of the Simplex algorithm requires a reduction of the pivot to 1, candidates were not penalised for using integer values in all their working.

Part (d) did not require the Simplex method. Candidates who used the three probabilities p, q and $(1 - p - q)$ for Bharveen were usually able to complete the question successfully.

Q24.

This simplex question was answered with more success than on previous papers. A number of candidates worked with integer values throughout and scored full marks. Some candidates failed to realise that the negatives in the profit row indicated that an optimum solution had not been found.

Q25.

A significant number of candidates failed to answer part (a) correctly, by writing that the functions exceeded the stated value rather than that this number was the maximum value for the function. Part (b) was well answered by the majority of candidates. Candidates who failed to score full marks did score some method marks but there were numerical slips when dealing with fractions. Candidates whose working used integers and only their final answers involved fractions were not penalised. A handful of candidates performed the pivot only once and ignored the fact that there were negative values on the 'cost line', but wrote down the values that they had at this stage.

Q26.

The majority of candidates scored the method marks on this question, but many numerical slips were made when dealing with fractions. Candidates were not penalised where all working used integers and only their final answer involved fractions; in fact candidates who used this method were very successful. Some candidates did not realise that the final 'cost line' should all have been positive to have reached an optimal solution.

Q27.

The majority of candidates scored the method marks on this question, but there were many numerical slips in dealing with fractions. Candidates will not be penalised if all their working uses integers and only their final answers involve fractions. Some candidates chose the wrong letter to pivot about at the start of the question and then didn't recognise the problem of having negatives in their 'cost line', and merely wrote down the value that they had for P.

Q31.

The figures in the tableau were chosen to keep the calculations relatively straightforward, and this certainly benefited most candidates. Furthermore, a pleasing number of candidates were able to interpret the figures.

Q33.

This was a fairly straightforward question on the simplex method. The most common and disappointing mistake was to choose the wrong pivoting point. Some candidates seemed unaware of the necessity to compare 26/2, 36/4 and 10/1 and choose the lowest.

Q35.

Candidates made very good attempts at this question and showed a pleasing understanding of the principles involved. They even coped surprisingly well with the arithmetic of the row operations required in the simplex method. The most common error was in part (c)(i) where candidates could read off the values of s and u from the given tableau, but failed to realise that $t = 0$ at that point.

Q38.

The vast majority of candidates simply solved the problem graphically, thus reducing the question to a much more straightforward one and missing out parts (a), (b) and (d) completely.

Q39.

Candidates using the simplex tableau in this question generally did better than those using the simplex method. This is a recurrent theme and candidates should be aware that the tableau approach is preferable.

Q40.

The graphical solution in part (a) of this question was very well done, with many candidates obtaining full marks. Far fewer candidates could apply an iteration of the simplex algorithm in part (b) correctly, although some excellent solutions were seen. Few candidates could identify the point on the graph that corresponded to the result of the iteration. More candidates knew that the slack variables would both have value zero at the end.

Q41.

Most candidates scored well on this question. Not all formulations were complete - “maximise” was often omitted. There was often no justification given for choosing 450 chairs and 10 tables. In part (c) good explanations of slack variables were rarely seen. In part (d) most candidates saw the obvious practical problem, but quite a few did not.